Hydrothermal Alteration and Vein Paragenesis of the Jiaodong Gold Province

Table 1. Electron microprobe analyses of pyrite and arsenopyrite.

Sample	Mineral	Fe (wt%)	S	As	Au (ppm)	Ag	Total
JD-01	Py-I	46.00	50.90	1.25	15.9	3.20	99.55
JD-02	Py-I	46.20	50.60	1.65	19.4	3.60	99.60
JD-03	Py-I	46.40	50.30	2.05	22.9	4.00	99.65
JD-04	Py-I	46.60	50.00	2.45	26.4	4.40	99.70
JD-05	Py-I	46.80	49.70	2.85	29.9	4.80	99.75
JD-06	Apy-I	47.00	49.40	3.25	33.4	5.20	99.80
JD-07	Apy-I	47.20	49.10	3.65	36.9	5.60	99.85
JD-08	Apy-I	47.40	48.80	4.05	40.4	6.00	99.90
JD-09	Apy-I	47.60	48.50	4.45	43.9	6.40	99.95
JD-10	Apy-I	47.80	48.20	4.85	47.4	6.80	100.00

2. Ore Fluid Evolution and Precipitation Mechanisms

Sulfidation and fluid immiscibility played dominant roles in gold precipitation across the deposit.

APPENDIX 1: Spectrometer calibration conditions and detection limits for EMPA standard calibrations.

Element	Line	Crystal	Standard Mineral	Peak (s)	Bkg (s)	LOD (ppm)
Fe	Ka	LIF	Natural Pyrite	20	10	120
S	Ka	PET	Natural Pyrite	20	10	150
As	La	TAP	Synthetic GaAs	30	15	250
Au	La	LIF	Pure Metal (99.99%)	40	20	180
Ag	La	PET	Pure Ag Metal	30	15	140
Cu	Ka	LIF	Natural Chalcopyrite	20	10	110
Zn	Ka	LIF	Natural Sphalerite	20	10	130
Pb	Ma	PET	Natural Galena	30	15	300
Co	Ka	LIF	Pure Co Metal	30	15	95
Ni	Ka	LIF	Natural Pentlandite	30	15	90
Sb	La	PET	Natural Stibnite	30	15	160
Bi	Ma	PET	Synthetic Bi2S3	40	20	280
Te	La	PET	Pure Te Metal	30	15	190
Se	Ka	LIF	Synthetic Bi2Se3	30	15	175

LOD = lower limit of detection calculated at 3· above background.